it's not unheard of for a codebase with few tests to lack written requirements; the acceptance tests act as baseline requirements as you move forward.

The general plan here is to use integration testing to quickly write high-level tests surrounding both the bug and the correct behavior it is related to—not to try to cover the entire application, but rather to cover the entire feature to discover regressions.

Acceptance-level tests are relatively easy to write for a legacy application but have limited utility. An integration test won't tell you where in the application you need to make the change that fixes the bug or adds the feature. Also, integration tests tend to run slowly, so you don't want them to be the only part of your test arsenal.

White-Box Testing

There's no way around writing real unit tests. Two distinct kinds of user tests are helpful when dealing with legacy applications. I'll talk about the standard TDD tests in a moment; first let's examine a technique that you can use to build up unit tests in hard-to-understand legacy code. You can use tests to figure out what is going on in the application—they are sometimes called *characterization tests*.

The basic process is straightforward: the difficulty of implementation depends on just how tangled the code is. First, select a method to test. Ideally the method should be related to a change you're planning to make in the app, although this process also works for "what is going on here?" exploration.

For this method, let's write a test we know will fail. You don't need to go deep into the internals for this. We're trying to behave like sonar—sending a test into the depths of the code and hoping to get a signal back:

```
it "calculates sales tax" do
  user = User.create(state: :il)
  order = user.orders.create
  order.line_items.create(:price => 250)
  order.line_items.create(:price => 300)
  expect(order.sales_tax).to eq(-300_000)
end
```

You have a simple, straightforward test, right up until the last line. (Insert cheesy DJ record-scratch sound effect.) You don't really expect the sales tax to be –300,000 dollars. But you don't want to guess what it is; you'll let the app tell you.